PORTFOLIO CASE STUDY · EXCEL

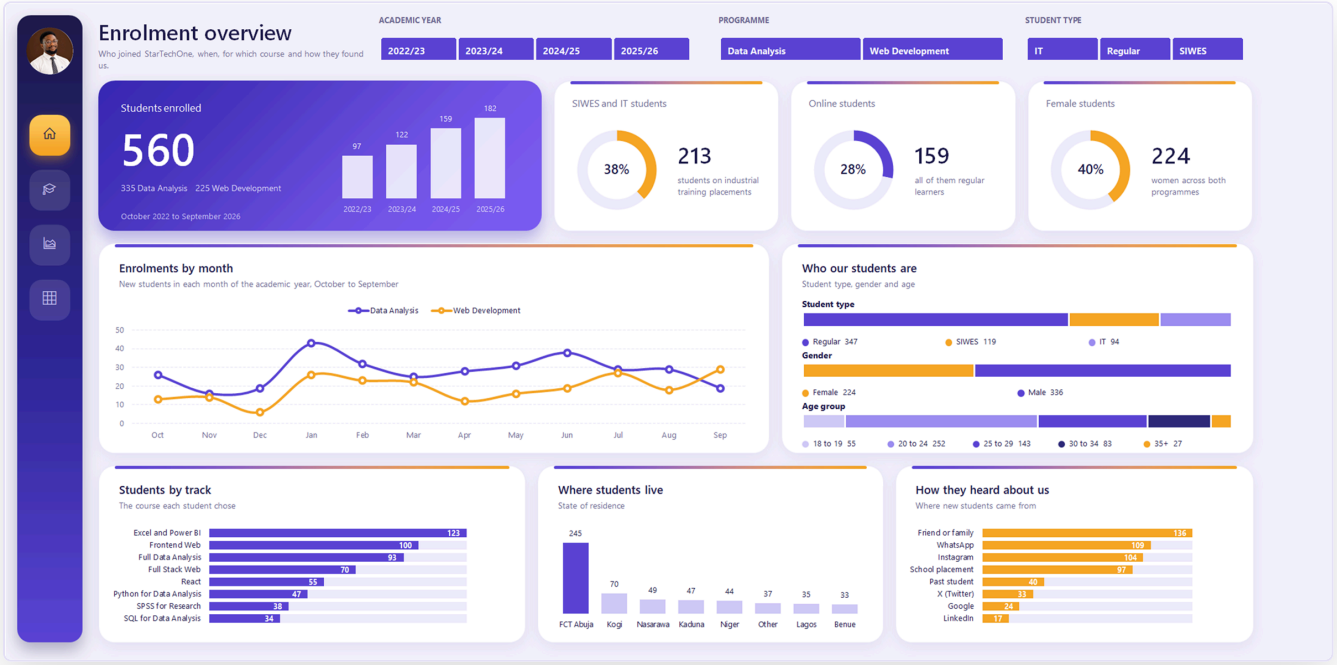
StarTechOne student enrolments dashboard

A two-page interactive Excel dashboard for a tech training academy: who joins, who finishes, and how much of the fees are collected, filtered with three clicks and no formulas.



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DOMAIN
Education and training

DATA
560 students, Oct 2022 to Sep 2026, 30 columns

MODEL
Excel Table, 14 PivotTables on one cache, formula layer

REPORT
2 dashboard pages, 2 analysis sheets, data dictionary

Excel · PivotTables · Slicers · GETPIVOTDATA · Dashboard design · Storytelling

THE PROJECT

The challenge

StarTechOne runs two programmes, Data Analysis and Web Development, for paying learners and for university and polytechnic students on placements. Management needed one place to see intake trends, learning outcomes and fee collection, and to filter all of it by year, programme and student type without touching a formula.

The data

A synthetic table of 560 student records (25 raw columns) built for portfolio use: enrolment, demographics, delivery mode, fees, acquisition channel, attendance, scores, grades, ratings and certificates. Five calculated columns add the academic year (October to September), month, age group, amount outstanding and certificate year.

My approach

1. Held the data in one Excel Table with five structured-reference calculated columns.
2. Built 14 PivotTables on a single shared cache, each shaped for one chart or KPI.
3. Added a GETPIVOTDATA formula layer so charts read stable ranges, never pivots directly.
4. Connected three slicers (Academic year, Programme, Student type) to every pivot on both pages.
5. Designed two card-based dashboard pages with a navigation rail, rings and linked text.

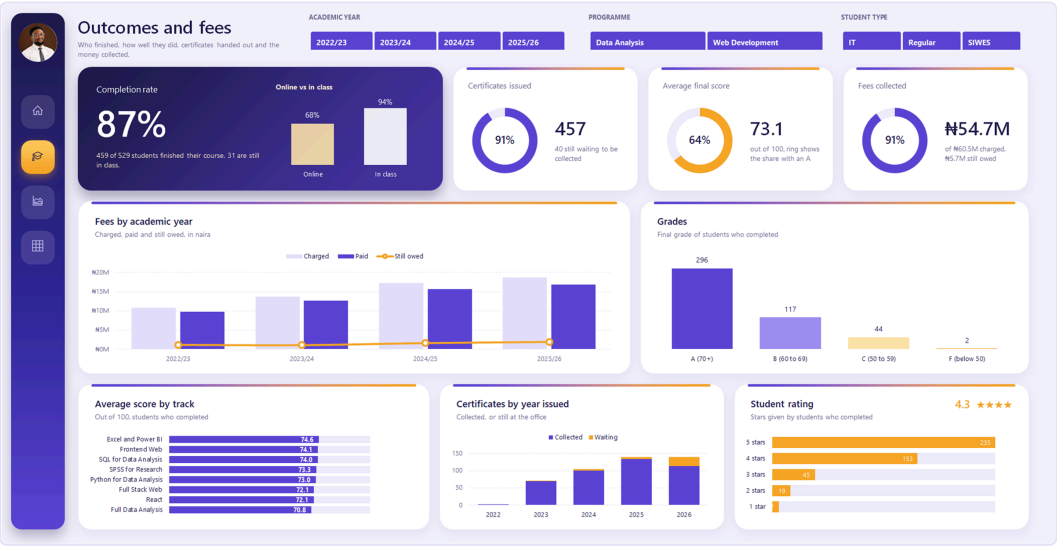
Headline numbers

560
students enrolled (+88% in four years)

87%
completion rate

68%
online completion vs 94% in class

91%
of N60.5M fees collected



Page 2 of the dashboard: completion, certificates, scores and fees.

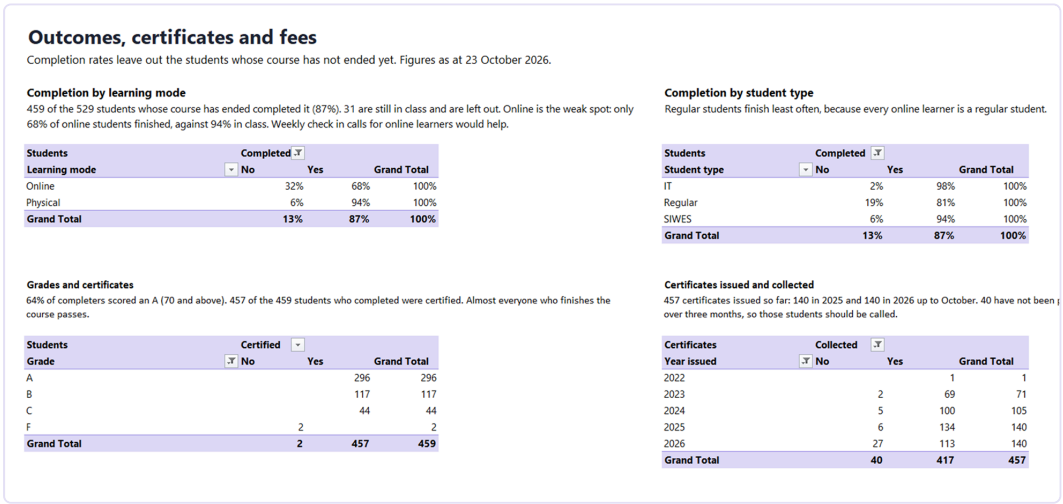
RESULTS

Key insights

Intake has nearly doubled	97 students in 2022/23 to 182 in 2025/26; Web Development grew every year.
Online learning is the weak spot	68% of online students finish against 94% in class.
Attendance predicts dropout	Completers attended 88% of classes; students who dropped out attended 50%.
Those who finish do well	Average score 73.1, 64% earned an A, and 457 of 459 completers were certified.
Fees are mostly collected	N54.7M of N60.5M paid; N3.3M of the N5.7M owed sits with non-completers.

Recommendations

1. Add weekly check-in calls for online learners.
2. Flag low attendance in the first weeks as an early warning.
3. Follow up part-paid fees before students drop out, and call the 19 students whose certificates have waited over three months.



Outcomes, certificates and fees

Completion rates leave out the students whose course has not ended yet. Figures as at 23 October 2026.

Completion by learning mode

459 of the 529 students whose course has ended completed it (87%). 31 are still in class and are left out. Online is the weak spot: only 68% of online students finished, against 94% in class. Weekly check in calls for online learners would help.

Students	Completed		
Learning mode	No	Yes	Grand Total
Online	32%	68%	100%
Physical	6%	94%	100%
Grand Total	**13%**	**87%**	**100%**

Completion by student type

Regular students finish least often, because every online learner is a regular student.

Students	Completed		
Student type	No	Yes	Grand Total
IT	2%	98%	100%
Regular	19%	81%	100%
SIWES	6%	94%	100%
Grand Total	**13%**	**87%**	**100%**

Grades and certificates

64% of completers scored an A (70 and above). 457 of the 459 students who completed were certified. Almost everyone who finishes the course passes.

Students	Certified		
Grade	No	Yes	Grand Total
A		296	296
B		117	117
C		44	44
F	2		2
Grand Total	**2**	**457**	**459**

Certificates issued and collected

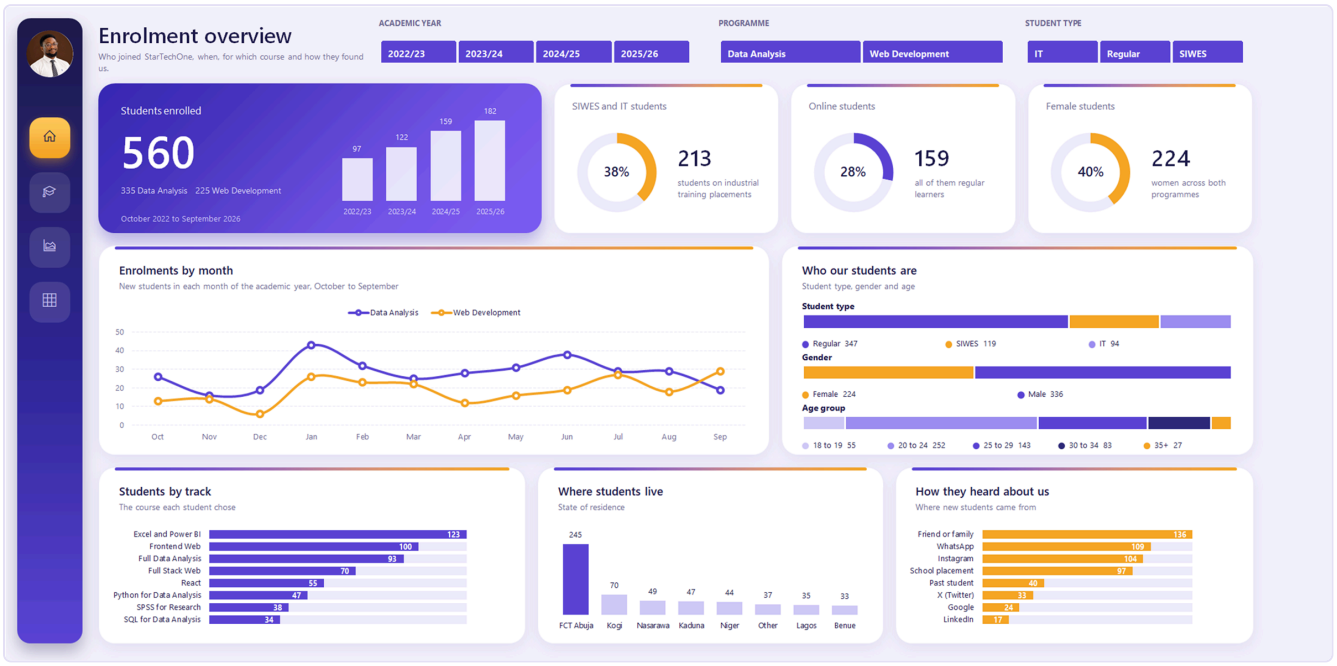
457 certificates issued so far: 140 in 2025 and 140 in 2026 up to October. 40 have not been p over three months, so those students should be called.

Certificates	Collected		
Year issued	No	Yes	Grand Total
2022		1	1
2023	2	69	71
2024	5	100	105
2025	6	134	140
2026	27	113	140
Grand Total	**40**	**417**	**457**

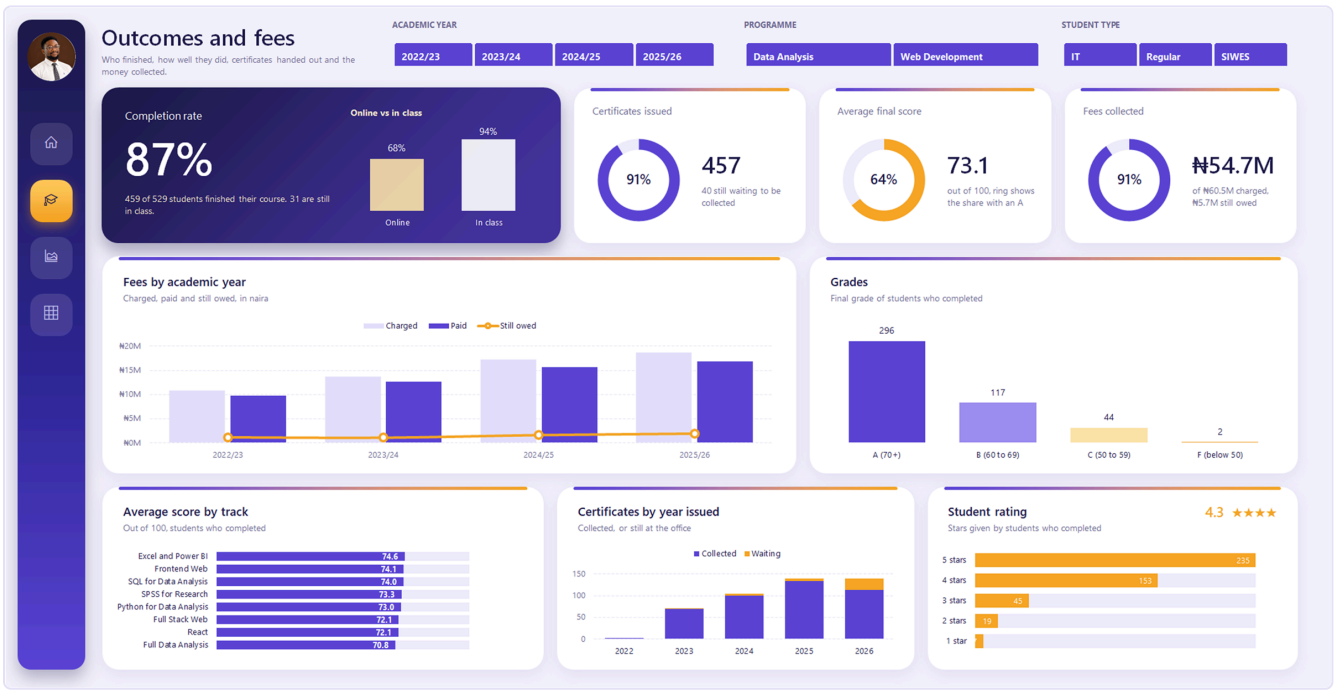
Part of an analysis sheet: each supporting table has a written takeaway above it.

A two-page interactive dashboard

Three slicers filter every number on both pages; a navigation rail moves between them; KPI text is linked to cells, so it updates with the charts.



Enrolment Dashboard. Intake by year, the student mix, tracks, geography and acquisition channels



Outcomes Dashboard. Completion, certificates, scores, grades, ratings and fees

QUALITY AND SKILLS

What I fixed, and what this shows

Design choices that keep the numbers right

- Completion leaves out the 31 students still in class, so the rate is not understated.
- IFERROR(GETPIVOTDATA(...), 0) shows zero, not #REF!, when a filter leaves no students.
- Charts read a formula layer, so PivotChart buttons and layout shifts never appear.

Skills demonstrated

- Data modelling in Excel Tables with calculated columns
- PivotTables on one cache with slicers across sheets
- A GETPIVOTDATA formula layer that decouples charts from pivots
- Dashboard design: grid, colour system, custom slicer style, navigation
- Analytical storytelling: findings and recommendations
- Documentation: data dictionary, metric definitions, limits

Tools

- Microsoft Excel (desktop)
- Excel Tables and structured references
- PivotTables, slicers and GETPIVOTDATA
- Line, bar, column, stacked and doughnut charts

Deliverables

- Interactive Excel workbook (2 dashboard pages)
- Full report PDF (every sheet)
- Detailed project guide

Column	What it means
About this data	Practice dataset of StarTechOne student enrolments, October 2022 to September 2026, 560 students. Figures are as at 23 October 2026. Made up for a portfolio dashboard; not real student records.
StudentID	Unique code for each student (STO0001 and so on), in order of enrolment.
EnrolmentDate	The day the student enrolled.
Programme	Data Analysis or Web Development.
Track	The course taken within the programme, for example Excel and Power BI, or Full Stack Web Development.
StudentType	Regular (paying learner), SIWES (university students on industrial work experience) or IT (polytechnic students on industrial training).
Status	Undergraduate, polytechnic student, graduate or working professional.
Gender	Female or Male.
Age	Age in years at enrolment.
State	State the student lives in.
LearningMode	Physical (in class) or Online. Only regular students can learn online; SIWES and IT students attend in person.
DurationWeeks	Length of the course or the training placement in weeks.
FeeNGN	Fee for the course in naira.
AmountPaidNGN	How much the student actually paid.
PaymentStatus	Paid in full, Part paid or Not paid.
HeardAboutUs	How the student found StarTechOne, for example Instagram, WhatsApp or school placement.
Attendance	Share of classes attended.
Completed	Yes if the student finished the course, No if they dropped out, In progress if their course has not ended yet.
FinalScore	Final assessment score out of 100. Blank if the student did not complete.
Grade	A (70 and above), B (60 to 69), C (50 to 59), F (below 50). Blank if not completed.
Certified	Yes if the student completed and scored 50 or more.
Rating	The student's rating of the course, 1 to 5 stars. Blank if not completed.
CourseEndDate	The day the course or training placement ends.
CertificateNo	Certificate number, given in the order certificates were issued. Blank if no certificate.
CertificateDate	The day the certificate was issued.
CertificateCollected	Yes if the student has picked up the certificate, No if it is still with us.

The data dictionary defines every column and states that the data is synthetic.

All data in this project is synthetic and was created for portfolio use.